

Derivative Pricing

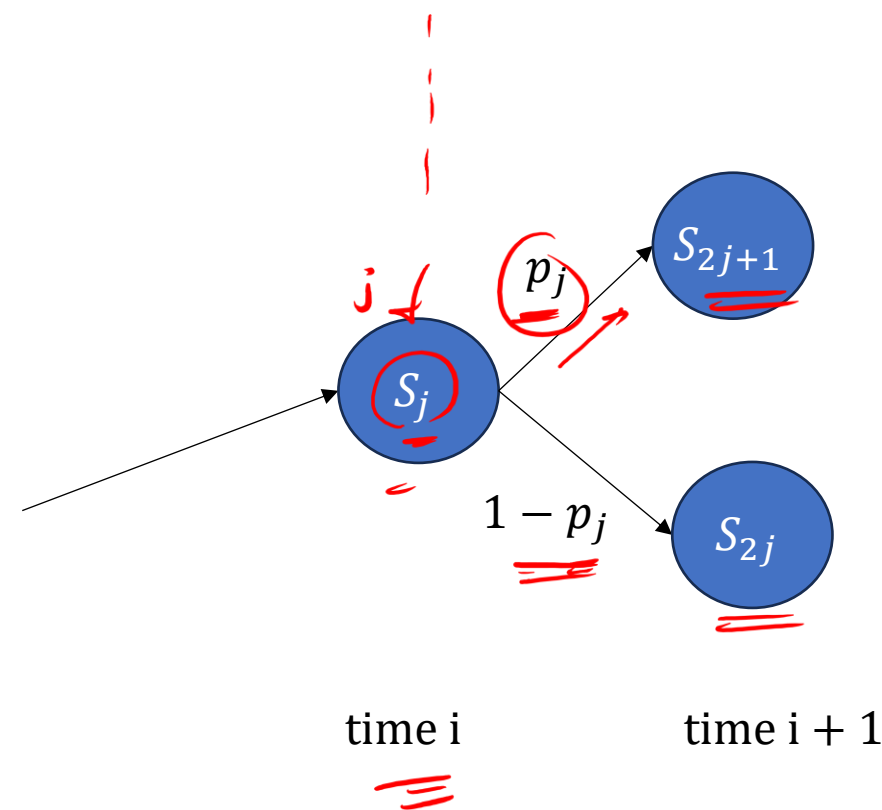
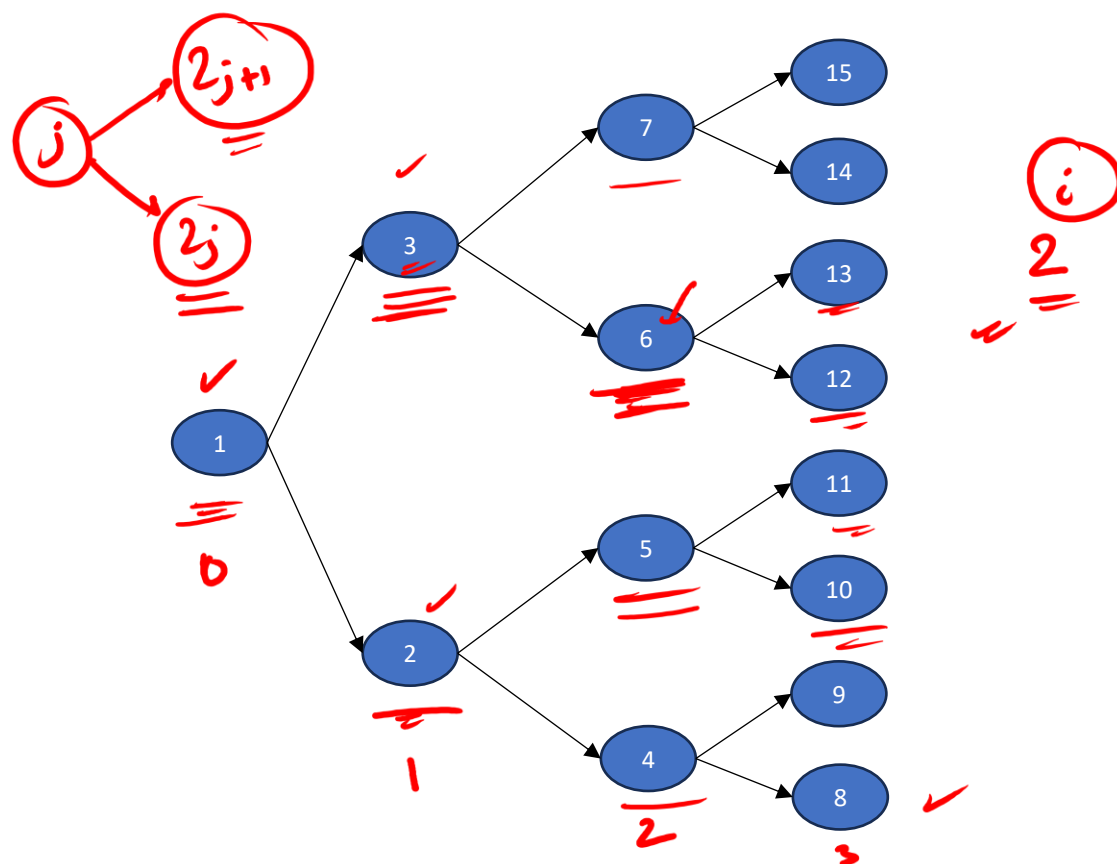
(2. The Binomial Tree Model)

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I. The two period Binomial Model

From Branch to Tree

- A one period model is a very basic infrastructure to do real world pricing as it only involves two states. It's a good starting point to understand the concepts. In real life, the stock price has infinitely many states. The ideal way to model would be a continuous model. However, we can use multi-period binomial model to get closer to reality.

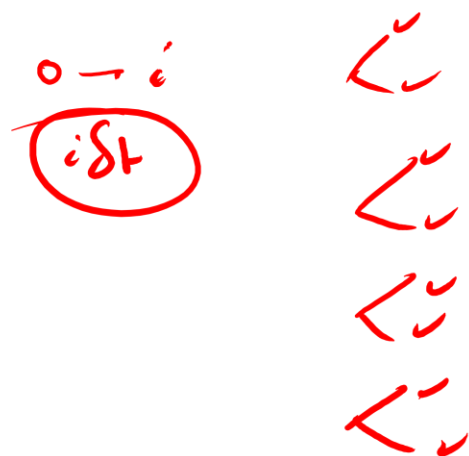




Stock and Cash account Dynamics

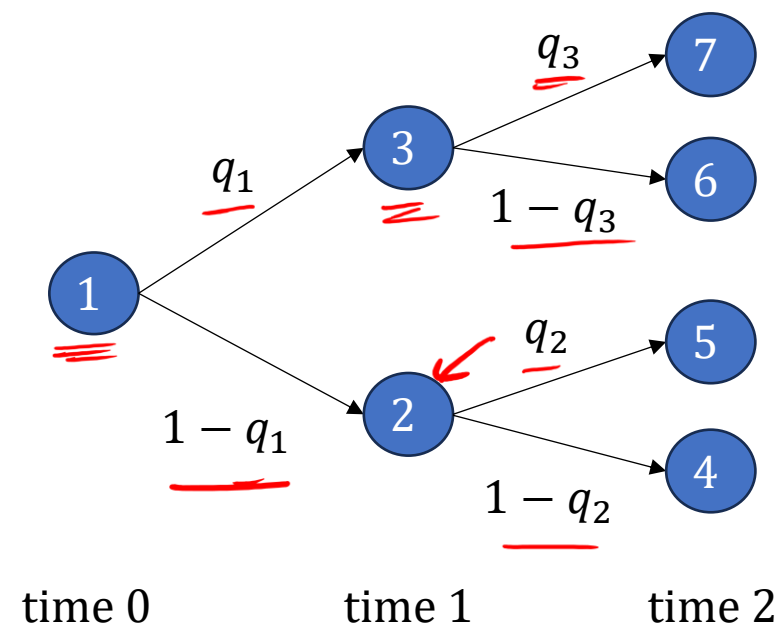
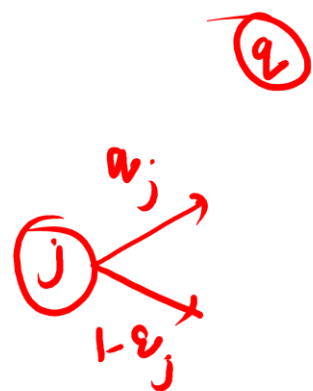
So, we will assume that each period is $\delta t = T/n$ where total time T has been broken into n time periods. We use index i to represent time and j to denote position of the node. At any time i , there are 2^i number of possible states of the stock price. But given the stock price at time $(i - 1)$, there are only two possible states; from node j it can go to either $2j$ or $2j + 1$

The cash account grows deterministically with risk-free rate $= r$. It's value at any node after time i is given as $B_i = B_0 e^{r(i\delta t)} = e^{r(i\delta t)}$ (Since $B_0 = 1$)



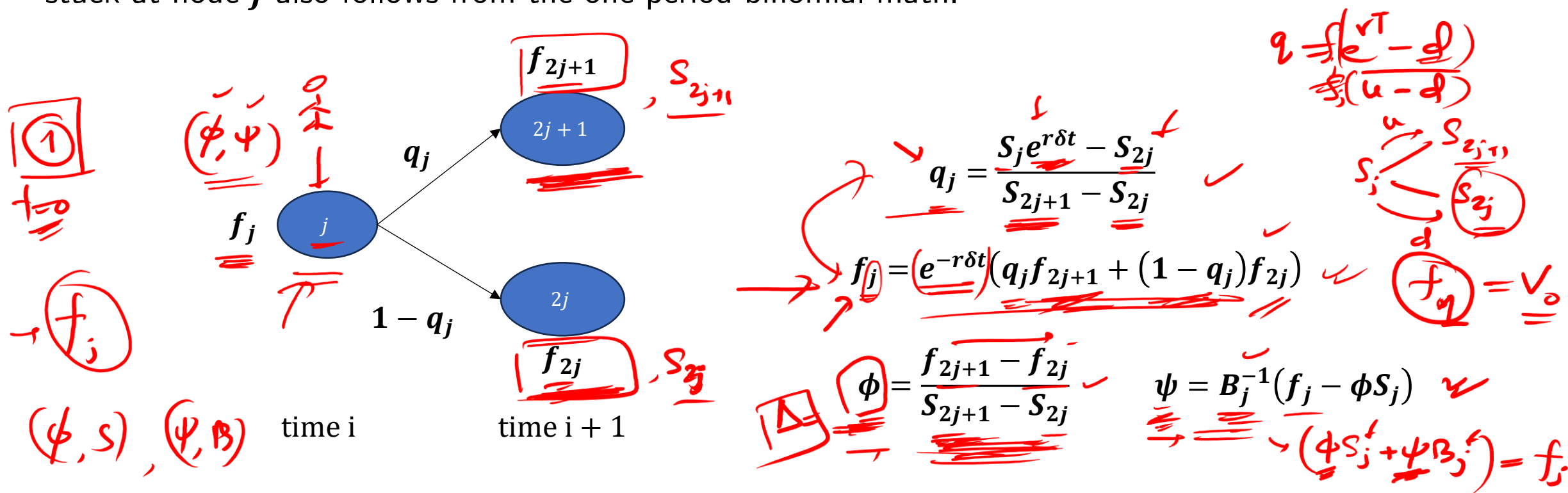
The 2-step Binomial (I)

Let's consider a 2-step Binomial Tree, so $n = 2$. We will replace the real-world probabilities with the risk-neutral ones for pricing exercise. At every-node j there exists the risk neutral probability q_j



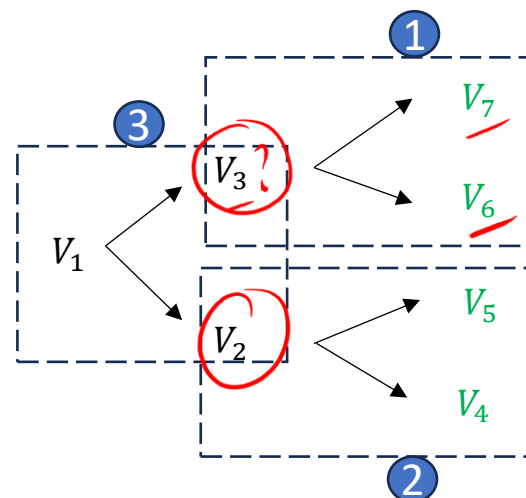
The 2-step Binomial (II)

Now the price of a derivative at any node j for time i given the values in time $(i + 1)$ is given by the discounted expectation under risk neutral probabilities as below. The replicating portfolio $\pi(\phi, \psi)$ to be stuck at node j also follows from the one period binomial math.

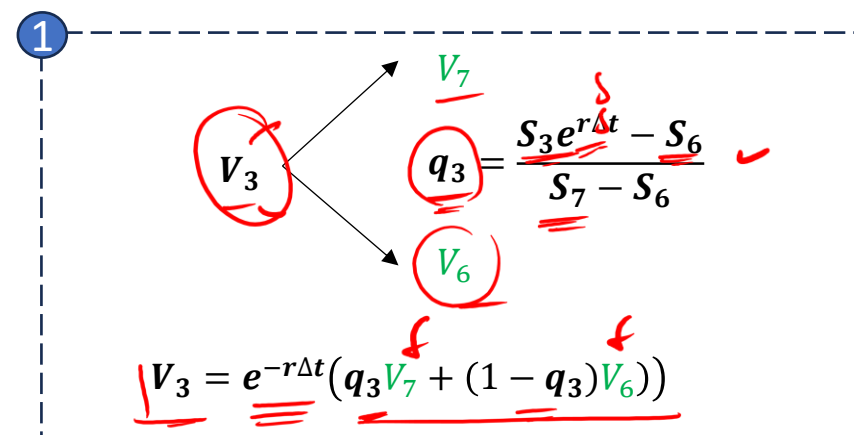


To solve for the derivative price at $t=0$, we proceed from right to left using **backward induction**.

Backward Induction



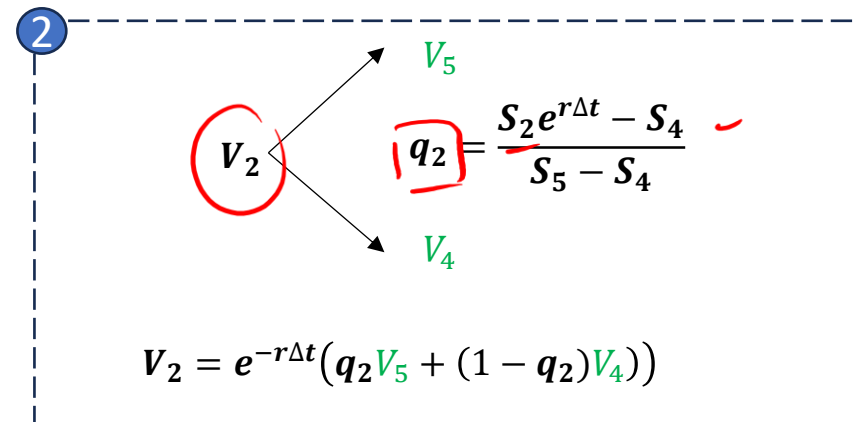
1



$$q_3 = \frac{S_3 e^{r\Delta t} - S_6}{S_7 - S_6}$$

$$V_3 = e^{-r\Delta t} (q_3 V_7 + (1 - q_3) V_6)$$

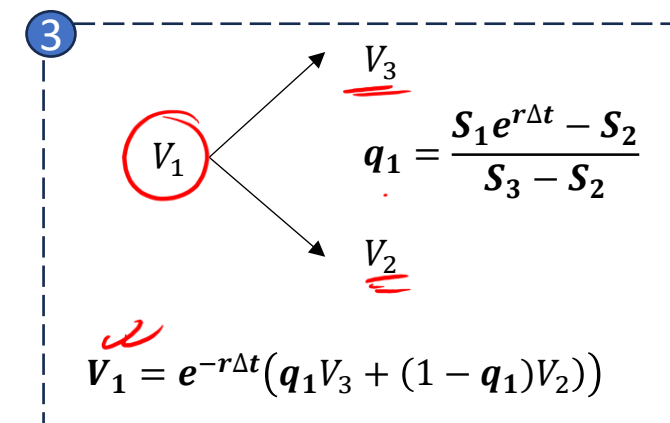
2



$$q_2 = \frac{S_2 e^{r\Delta t} - S_4}{S_5 - S_4}$$

$$V_2 = e^{-r\Delta t} (q_2 V_5 + (1 - q_2) V_4)$$

3



$$q_1 = \frac{S_1 e^{r\Delta t} - S_2}{S_3 - S_2}$$

$$V_1 = e^{-r\Delta t} (q_1 V_3 + (1 - q_1) V_2)$$

Example

Consider a two-period binomial tree model for a stock with the following details

Current stock price = 40

In the first period, the stock may go up to 60 or go down to 30

Following the up-move in the 1st period, the stock can go up to 80 or go down to 50

Following the down-move in the 1st period, the stock can go up to 40 or go down to 25

The real-world probabilities are $\frac{1}{2}$ for all periods (~~X~~)

Continuously compounded risk-free rate = 5% per year

$$\max(S_T - K, 0) = X$$

$$(S_T - K)^+$$

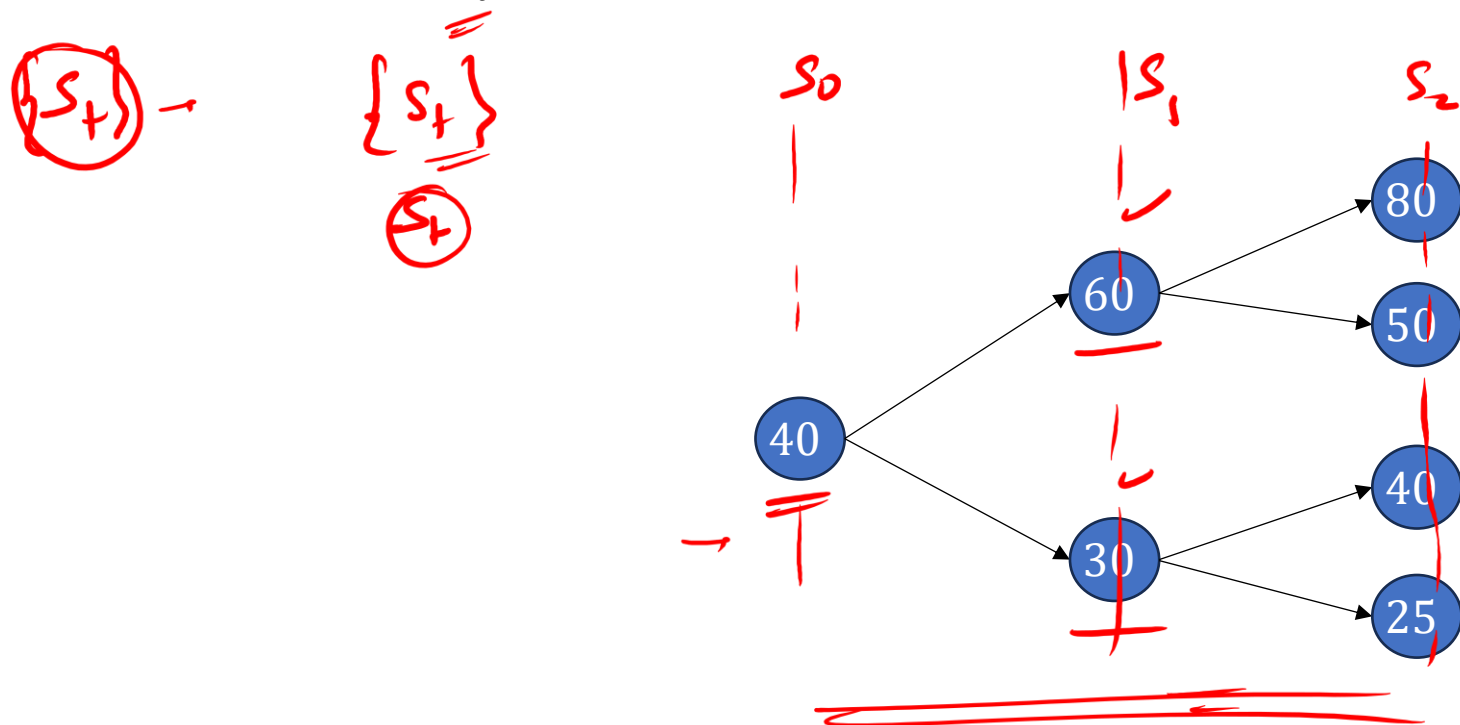
- Find the price of a European Call option maturing in 2 years that has exercise price of 45 using expectation under risk-neutral probability. Find the composition of the replicating portfolio.
- Find the price of a European Put option maturing in 2 years that has exercise price of 45
- Derive the put-call parity using no-arbitrage arguments $\underline{P_0 + S_0} = \underline{C_0 + Ke^{-rT}}$

II. The Se7en

1. Process

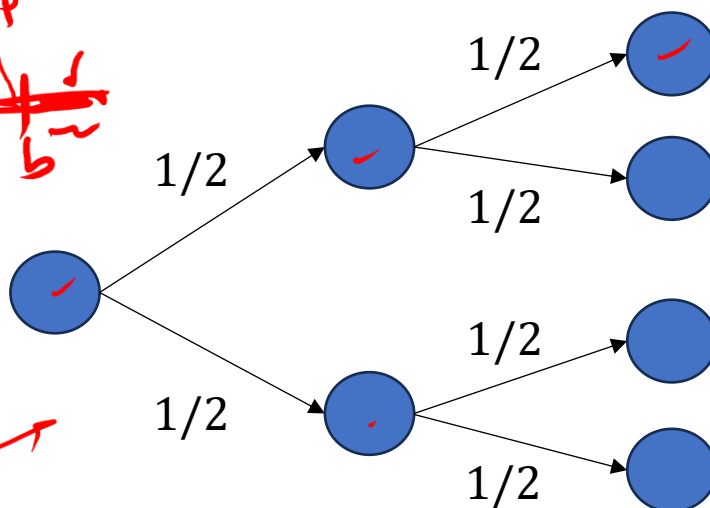
The stock price evolves with time and can take a set of possible values at a given time. The stock price is called the **process** (or stochastic process). For, example one possible process is given below

The random variable S_i denotes the value of the process at time i . For example, S_1 is either 60 or 30.

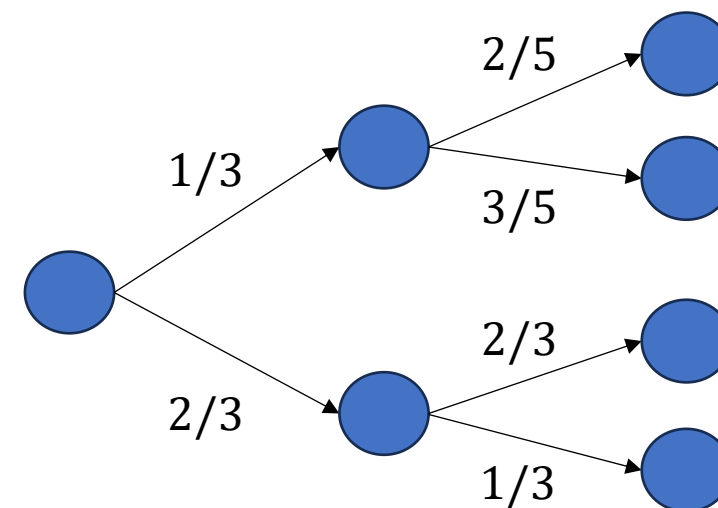


2. Probability

Probabilities are separate from the process. As we have learned so far, we have seen two **probability measures**; the **real-world** measure $\mathbb{P}$ and the **risk-neutral** measure $\mathbb{Q}$. The probability only tells us how likely are the up moves and down moves.



measure $\mathbb{P}$

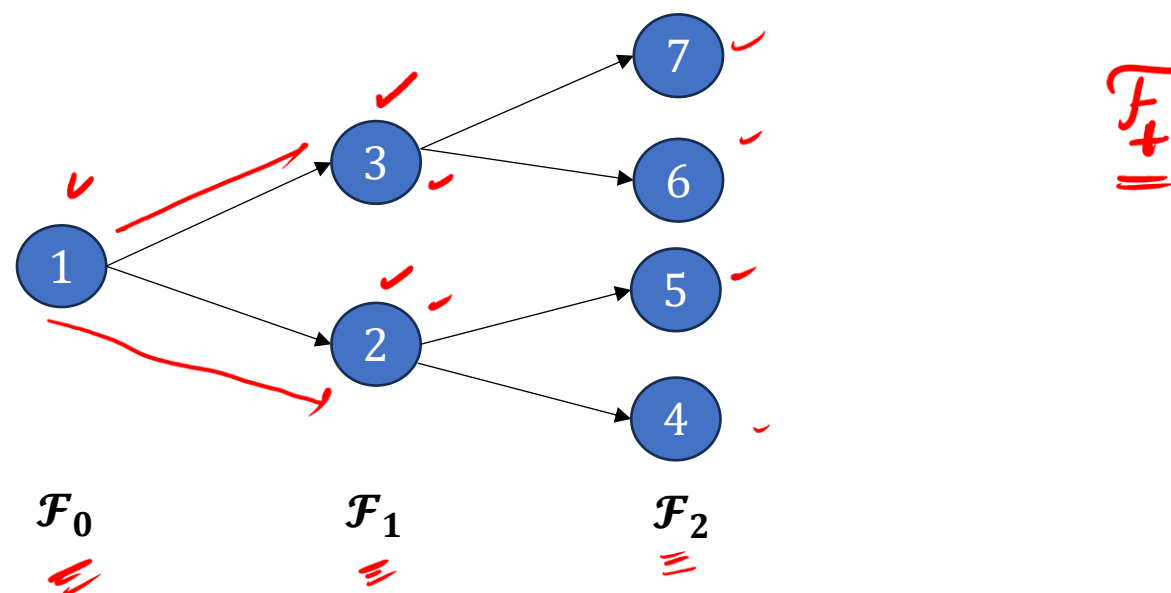


measure $\mathbb{Q}$

Two measures $\mathbb{P}$ and $\mathbb{Q}$ are called **equivalent** if they agree on the null sets; meaning if an event is impossible under $\mathbb{P}$, it must be impossible under $\mathbb{Q}$ as well.

3. Filtration

Filtration ($\mathcal{F}_i$) is the history of stock price until time i . The filtration starts with time 0 with $\mathcal{F}_0$ equal to the starting node 1, i.e., $\mathcal{F}_0 = \{1\}$. At time 1, the filtration is $\mathcal{F}_1 = \{\{1,2\}, \{1,3\}\}$ depending on whether the first jump was down or up. Similarly, $\mathcal{F}_2 = \{\{1,2,4\}, \{1,2,5\}, \{1,3,6\}, \{1,3,7\}\}$. Knowing the filtration pin-points the sample path history till that point. In other words, filtration tells us the exact node in time where we are right now.

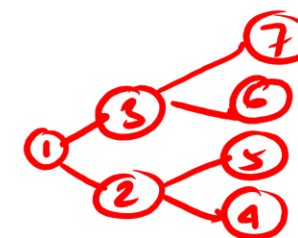
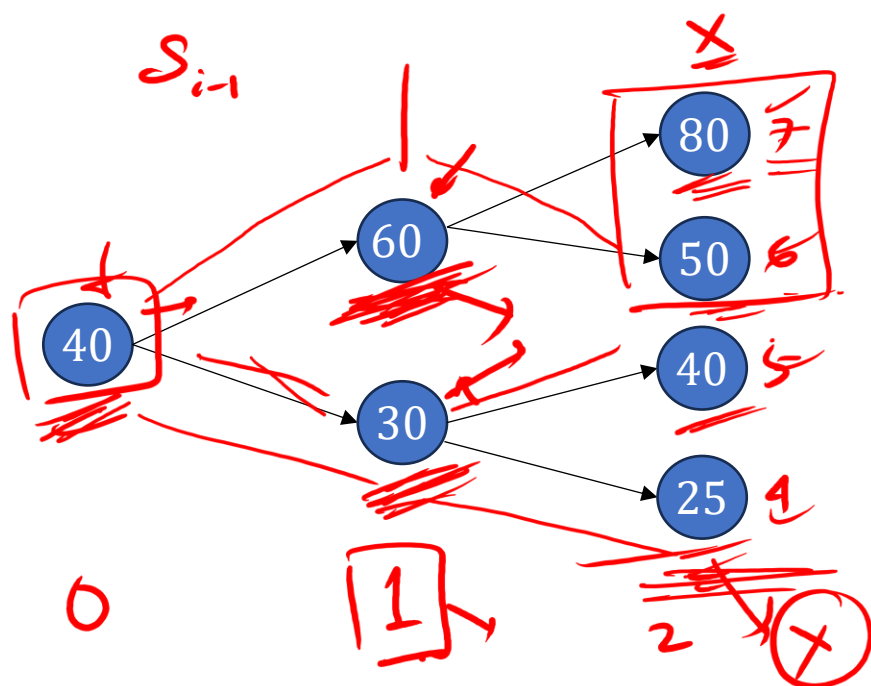


Technically, the Filtration will also contain all elements of previous filtration, but it's skipped for better readability and the sake of argument. (Refer to Math Primer for a detailed account)

4.Claim

A **Claim** (or contingent claim) X is a function the nodes of the tree until a time horizon (expiry) T . Or in other words, it's a function of $\mathcal{F}_T$. In layman terms, we are referring to payoff of a security. The key difference is that the Claim is only defined at T , whereas the process is defined at all times up-to T

Let's say we have a 2-period binomial tree for maturity T .



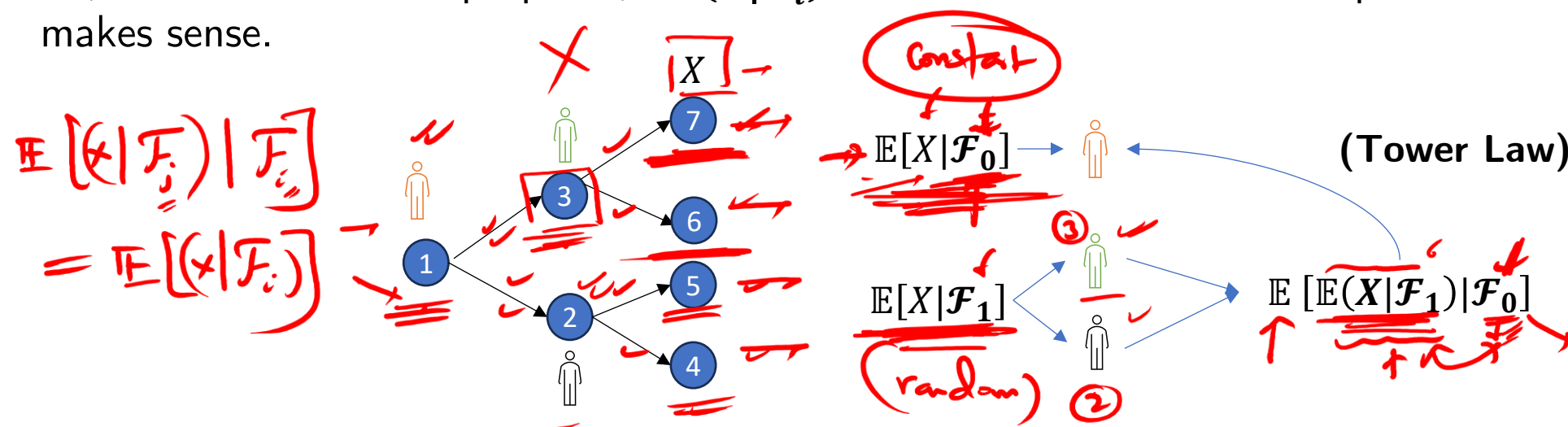
Node	S_2	$(S_2 - 45)^+$	$\max\{S_0, S_1, S_2\}$
7	80	35	80
6	50	5	60
5	40	0	40
4	25	0	40

5. Conditional Expectation

The **conditional expectation** $\mathbb{E}^{\mathbb{Q}}(\cdot | \mathcal{F}_t)$ operator incorporates two inputs

- Conditional on Filtration $\mathcal{F}_t$ which means information available till t
- Measure $\mathbb{Q}$ which gives the path probabilities

The conditional expectation $\mathbb{E}^{\mathbb{Q}}(X | \mathcal{F}_i)$ means we are teleported to one of the nodes in time i . And we are calculating expectation of X considering all paths emerging from that node. But standing at time 0 , this expectation $\mathbb{E}^{\mathbb{Q}}(X | \mathcal{F}_i)$, $i > 0$ is still random. Because, at time 0 , that future node is known yet. So, for all intents and purposes, $\mathbb{E}^{\mathbb{Q}}(X | \mathcal{F}_i)$ is a random variable and expressions like $\mathbb{E}^{\mathbb{Q}}[\mathbb{E}^{\mathbb{Q}}(X | \mathcal{F}_i) | \mathcal{F}_0]$ makes sense.

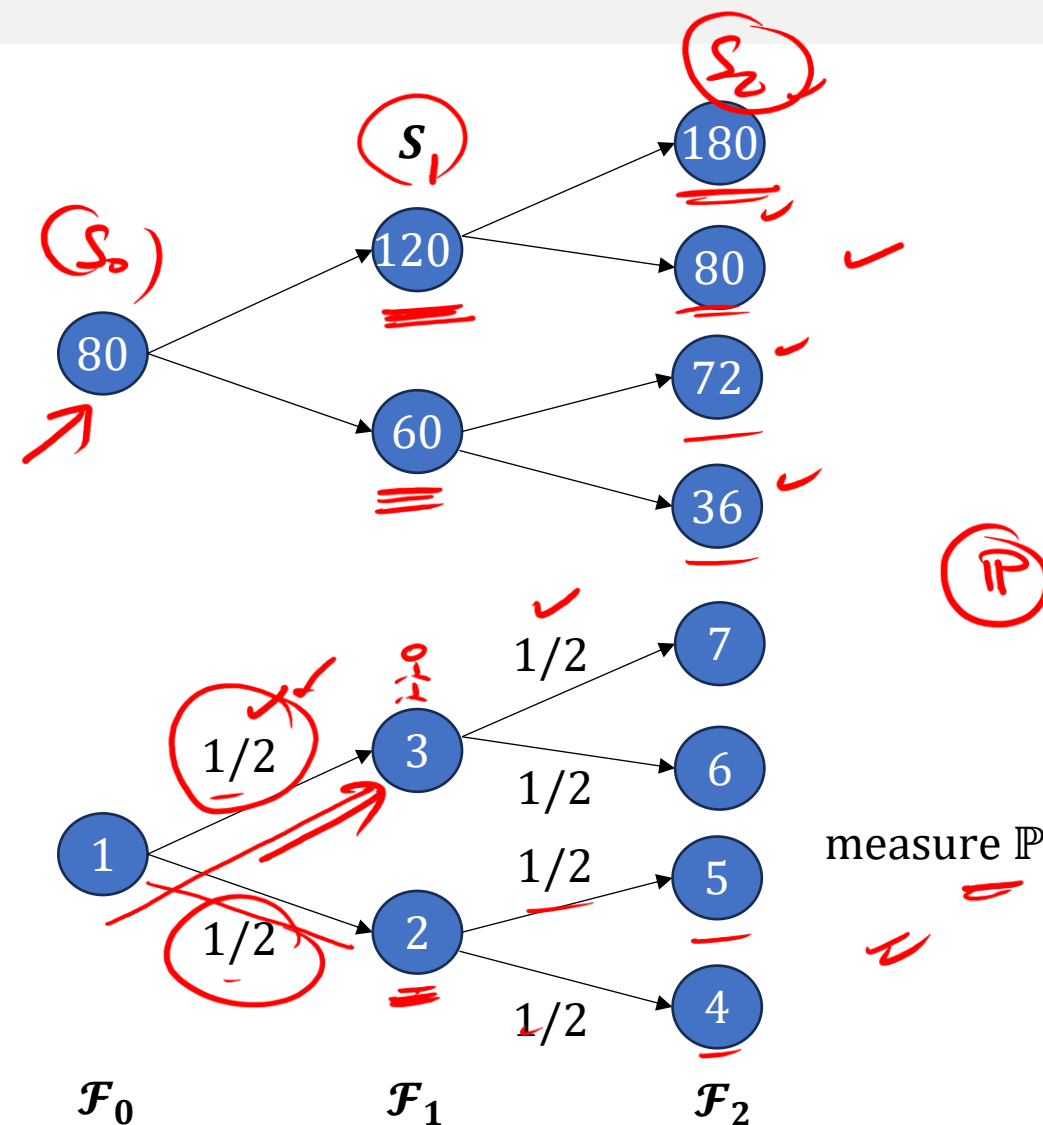


Examples

Calculate the following Expectations

Expectation	Filtration	Value
$\mathbb{E}^{\mathbb{P}}[S_2 \mathcal{F}_0]$	$\rightarrow \{1\}$	<u>92</u>
$\mathbb{E}^{\mathbb{P}}[S_2 \mathcal{F}_1]$	$\{1,2\}$	<u>54</u>
	$\{1,3\}$	<u>130</u>
$\mathbb{E}^{\mathbb{P}}[S_2 \mathcal{F}_2]$	$\{1,2,4\}$	<u>36</u>
	$\{1,2,5\}$	<u>72</u>
	$\{1,3,6\}$	<u>80</u>
	$\{1,3,7\}$	<u>180</u>

$$180 \times \frac{1}{4} + 80 \times \frac{1}{4} + 72 \times \frac{1}{4} + 36 \times \frac{1}{4} = 92$$

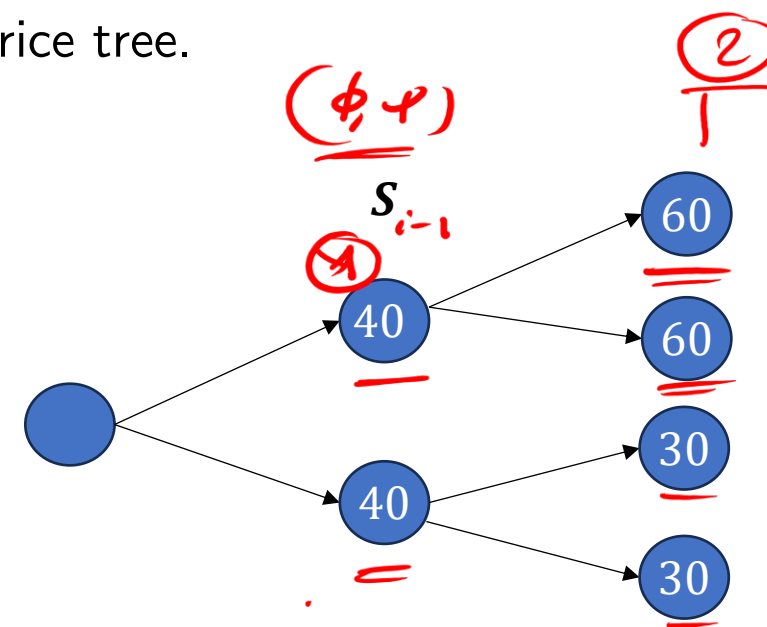


6. Previsible Process

A pre-visible process $\underline{\phi} = \underline{\phi}_i$ is a process on the same tree whose value only depends on one time step before time i , $\mathcal{F}_{i-1}$. This process follows its own binomial tree. But, at each node where S is known, ϕ is known one period in advance. We can say $\underline{\phi}_i$ is $\mathcal{F}_{i-1}$ measurable.

For example,

- the cash account price $\underline{B}_i$ is pre-visible on the original stock price tree.
- The process $\underline{S}_{i-1}$, $i \geq 1$ is also pre-visible.



Can you think of another process that is pre-visible process ? ✓

7. Martingale Measure

A process Y is a martingale under measure $\mathbb{P}$ and Filtration $\mathcal{F}_i$ if

$$\mathbb{E}^{\mathbb{P}}[Y_k | \mathcal{F}_i] = Y_i \quad \text{For all } i \leq k$$

Then we call Y as a $\mathbb{P}$ martingale.

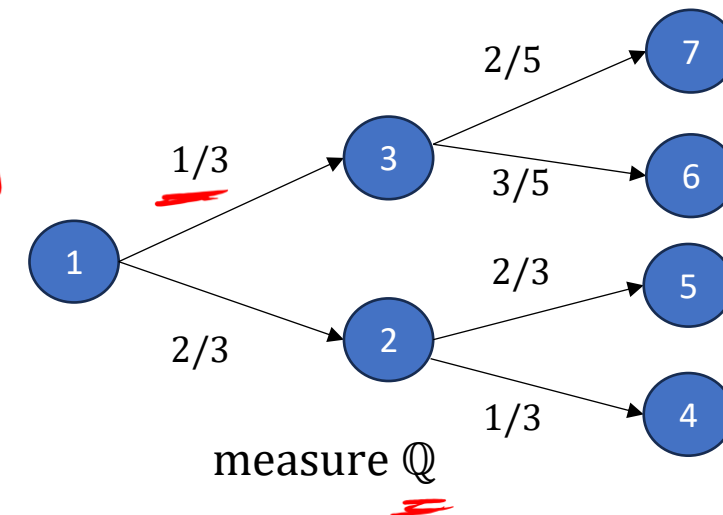
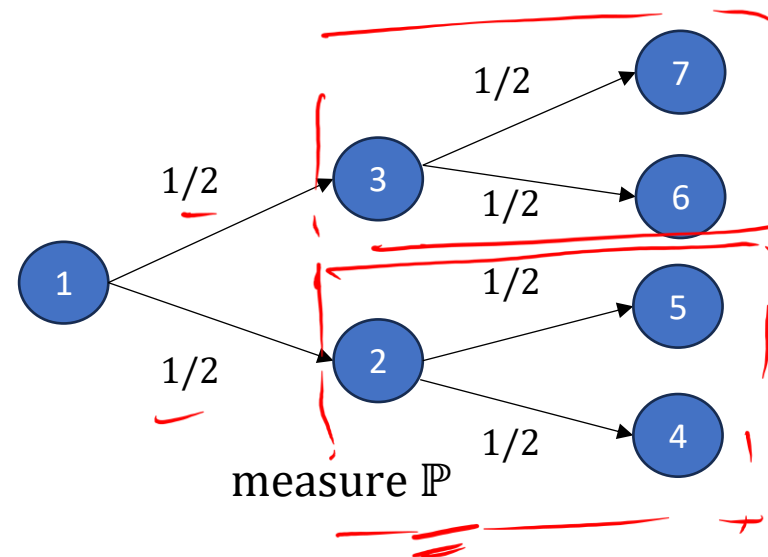
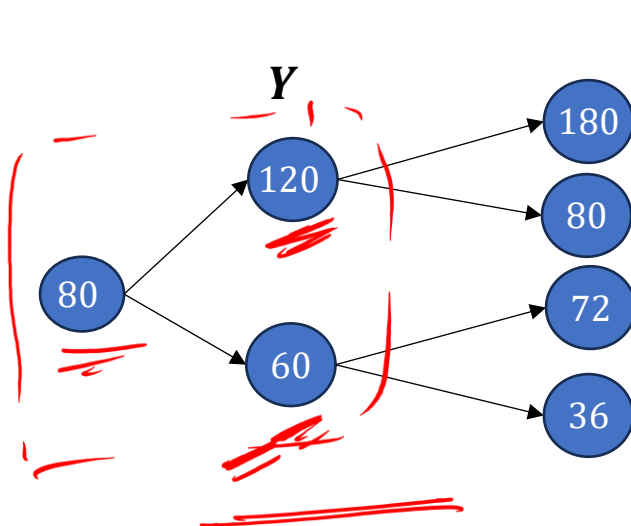
Examples :

- A **constant** process (say $Y = 100$) is trivially martingale under any measure.
- The **conditional expectation** under a specific measure is auto-martingale under that measure. This is because of Tower Law, $\mathbb{E}^{\mathbb{Q}}[\mathbb{E}^{\mathbb{Q}}(Y | \mathcal{F}_k) | \mathcal{F}_i] = \mathbb{E}^{\mathbb{Q}}(Y | \mathcal{F}_i)$. So, we can say that $\mathbb{E}^{\mathbb{Q}}(Y | \mathcal{F}_i)$ is a $\mathbb{Q}$ - martingale and $\mathbb{E}^{\mathbb{P}}(Y | \mathcal{F}_i)$ is a $\mathbb{P}$ - martingale.

$$\mathbb{E}^{\mathbb{P}}[\mathbb{E}^{\mathbb{P}}(Y | \mathcal{F}_k) | \mathcal{F}_i] = \mathbb{E}^{\mathbb{P}}(Y | \mathcal{F}_i)$$

Martingale Measure (Exercise)

Consider the following process Y and two possible measures $\mathbb{P}$ and $\mathbb{Q}$



Verify if the following are martingale

a) Y under $\mathbb{P}$

b) Y under $\mathbb{Q}$

$$\begin{aligned} \rightarrow \mathbb{E}^{\mathbb{P}}[Y_1 | \mathcal{F}_0] &= 120 \times \frac{1}{2} + 60 \times \frac{1}{2} = 90 \\ &= Y_0 = 80 \end{aligned}$$

$$\begin{aligned} 120 \times \frac{1}{3} + 60 \times \frac{2}{3} &= \\ 40 + 40 &= 80 \end{aligned}$$



Risk-Neutral Growth = Equivalent Martingale measure

It's time we ask the question, if there is anything special about the risk-neutral measure. Remember risk-neutral probabilities ensure the asset's expected growth is risk-free rate. But is there more ?

Define a process $\underline{Z}_i = \underline{B}_i^{-1} \underline{S}_i$ which is the discounted-stock process.

Verify if it's a martingale under $\mathbb{Q}$

$$\begin{aligned}
 \mathbb{E}^{\mathbb{Q}}[Z_{i+1} | \mathcal{F}_i] &= Z_i \\
 &= \mathbb{E}^{\mathbb{Q}}\left[\frac{S_{i+1}}{B_{i+1}} \mid \mathcal{F}_i\right] = e^{-r(i+1)\delta t} \cdot (q u S_i + (1-q) d S_i) \\
 &= S_i e^{-r(i+1)\delta t} \left(\underline{q(u-d)} + \underline{d} \right) \quad \underline{q} = \frac{e^{r\delta t} - d}{u - d} \\
 &= S_i e^{-r(i+1)\delta t} \cdot e^{r\delta t} = S_i e^{-r(i\delta t)} = \frac{S_i}{B_i} = Z_i
 \end{aligned}$$

Martingale Representation Theorem

Martingale representation theorem →

Let $\underline{S}$ be a binomial price process and it's a $\mathbb{Q}$ -martingale. $\underline{N}$ be another $\mathbb{Q}$ -martingale, then there exists a pre-visible process ϕ such that

$$\begin{aligned} \mathbb{E}^{\mathbb{Q}}[S_{i+1} | \mathcal{F}_i] &= S_i \\ \mathbb{E}^{\mathbb{Q}}[N_{i+1} | \mathcal{F}_i] &= N_i \end{aligned}$$

$$N_i = N_0 + \sum_{k=1}^i (\phi_k \Delta S_k)$$

Handwritten notes: ΔN_k above the sum, dN below the sum.



where, $\Delta S_k = S_k - S_{k-1}$. The theorem states we can construct a martingale process from $\underline{S}$ pre-visibly with steps known in advance. Note that Martingales have zero drifts, so the only way two martingales are different is through an initial offset and the width of up-state and down-state. ϕ plays the role of slope $\phi = \frac{\Delta N_k}{\Delta S_k}$ and the expression is discrete sum of ΔN_k from 0 to i .

$$\underline{y}_t = \underline{y}_0 + \sum (\text{slope} \times \underline{\Delta x})$$

Handwritten notes: Δy above the sum, Δx below the sum.

Pricing Application

How are these results and theorems interconnected when it comes to the pricing use case ? We know how to link two martingales process using “martingale representation theorem”.

We know discounted stock price $Z_i = B_i^{-1}S_i$ is martingale under $\mathbb{Q}$.

Consider the random variable **discounted claim** to time 0 i.e., $B_T^{-1}X$. Now we can create a process from the discounted claim, $(E_i) = \mathbb{E}^{\mathbb{Q}}(B_T^{-1}X | \mathcal{F}_i)$ which is an auto-martingale under $\mathbb{Q}$

Martingale representation theorem gives

$$E_i = E_0 + \sum_{k=1}^i (\phi_k) \Delta Z_k$$

$\Delta = \frac{\partial v}{\partial S}$ $(\Delta S) \rightarrow (\Delta v)$

Construction Strategy

Now at time tick i , in our replicating portfolio $\pi_i(\phi, \psi)$ we can hold

- ϕ_{i+1} units of Stock S_i
- $\psi_{i+1} = (E_i - \phi_{i+1}B_i^{-1}S_i)$ units of cash account B

$$\pi_i = \phi_{i+1}S_i + \psi_{i+1}B_i = \phi_{i+1}B_iE_i$$

At time zero, the portfolio $\pi_0 = \phi_1S_0 + \psi_1B_0 = E_0B_0 = \mathbb{E}^Q(B_T^{-1}X|\mathcal{F}_0)$, since $B_0 = 1$. This is the initial cost of setting up the replicating portfolio. Choosing (ϕ_1, ψ_1) at time zero is not a problem as they are pre-visible.

At time 1, the portfolio π_0 would become

$$\phi_1S_1 + \psi_1B_1 = B_1(E_0 + \phi_1(B_1^{-1}S_1 - B_0^{-1}S_0)) = B_1(E_0 + \phi_1(\Delta Z_1))$$

Which using martingale representation theorem, is B_1E_1 . At time 1, we need to reconstruct our new portfolio π_1 . Irrespective of which node we land in time 1, the cost of new portfolio π_1 is B_1E_1 . In other words, this portfolio is **self-financing** (no extra money is added or removed from the portfolio, all portfolio changes are from change in stock and cash positions only)

No Arbitrage Price

The replicating portfolio π_{T-1} at time $(T-1)$ would grow in the final time step to become

$$E_T = \mathbb{E}^Q \left(\frac{X}{B_T} \middle| \mathcal{F}_T \right) \quad \text{and} \quad B_T E_T = B_T (B_T^{-1} X) = X$$

Which is the final claim. Since the replicating portfolio matches the claim at T , it's price at time 0 must be the price of the derivative at time 0. In other words,

$$i \quad E_i B_i$$

$$(V_0) = E_0 = \mathbb{E}^Q (B_T^{-1} X | \mathcal{F}_0)$$

More generally, the time i price of the contingent claim X maturing at T

$$V_i = B_i \mathbb{E}^Q (B_T^{-1} X | \mathcal{F}_i) \rightarrow \left(\frac{V_i}{B_i} \right) = \mathbb{E}^Q \left(\frac{V_T}{B_T} \middle| \mathcal{F}_i \right)$$

Which also means **discounted derivative** price (V_i/B_i) is also a martingale under measure $\mathbb{Q}$

Important points to remember

From the work of Harrison and Pliska (1983)

$$X = \phi_1 S_1 + \psi_1 B_1 \leftarrow \begin{matrix} \alpha V_1 + \beta V_2 \\ Z^{\nu_1} \\ Z^{\nu_2} \end{matrix} \quad \begin{matrix} \uparrow \\ \nu_1 \\ \nu_2 \end{matrix}$$

- ▶ A contingent claim is **attainable** if there exists a self-financing strategy ϕ such that $\pi_T(\phi) = X$
- ▶ The market is **free of arbitrage** if (and only if) there exists a **martingale measure**;
- ▶ The market is **complete** if and only if the martingale measure is **unique**;
- ▶ In an arbitrage-free market, not necessarily complete, the price of any attainable claim is uniquely given, either by the value of the associated replicating strategy, or by the risk neutral expectation of the discounted claim payoff under any of the equivalent (risk-neutral) martingale measures.